

# Proofs

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Proofs are the primary means by which mathematical truth is established. The purpose of a proof is to justify the truth of a mathematical statement (called the **conclusion**) by considering other mathematical statements (called **premises**). Once a statement is proved, we refer to it as a **theorem**.

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- **Definition 1: Conjunctions ( $P \wedge Q$ )**

A statement of the form  $P \wedge Q$  is true when both  $P$  and  $Q$  are true. To prove a conjunction, it suffices to prove each of its component statements separately. That is, if we establish  $P$  and we also establish  $Q$ , we may conclude  $P \wedge Q$ .

- **Example 1.1**

To prove that a number  $n$  is even and greater than 10, we prove separately that  $n$  is an even number and that  $n > 10$ . Combining them yields the result.

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- **Definition 2: Disjunctions ( $P \vee Q$ )**

A statement of the form  $P \vee Q$  is true whenever at least one of its components is true. Therefore, to prove a disjunction, it is enough to prove one of its disjuncts (though a false disjunct cannot establish the disjunction).

- **Example 2.1**

To prove that a number  $n$  is either even or odd, it suffices to prove that  $n$  is even, and the disjunction follows.

- **Example 2.2**

Let  $a, b \in \mathbb{R}$  that satisfy  $a + b = 10$ . Prove that  $ab \leq 25$  or  $a^2 + b^2 < 10$ .

- **Solution 2.2.1**

Suppose  $a, b \in \mathbb{R}$  satisfy  $a + b = 10$ , then  $a = 10 - b$ , and we have

$$\begin{aligned} ab &= (10 - b)b \\ &= 10b - b^2 \\ &= -(b^2 - 10b) \\ &= -(b^2 - 10b + 25) + 25 \\ &= -(b - 5)^2 + 25 \\ &\leq 25 \end{aligned}$$

So  $ab \leq 25$ , then  $ab \leq 25$  or  $a^2 + b^2 < 10$ .

- **Example 2.3**

Prove that  $(\forall x) P(x) \vee (\forall x) Q(x) \implies (\forall x) (P(x) \vee Q(x))$  and refute the converse implication.

- **Solution 2.3.1**

Suppose  $(\forall x) P(x) \vee (\forall x) Q(x)$  is true, then  $(\forall x) P(x)$  is true or  $(\forall x) Q(x)$  is true.

Suppose  $(\forall x) P(x)$  is true, then  $\forall x, P(x)$  is true, then  $\forall x, P(x) \vee Q(x)$  is true and then  $\forall x, (P(x) \vee Q(x))$  is true.

To check that the converse is false, consider  $P(x) : x$  is even,  $Q(x) : x$  is odd.

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- **Definition 3: Indirect Strategy for Disjunction**

Another useful strategy to prove  $P \vee Q$  is to proceed indirectly. Assume that one of the disjuncts is false (say  $\neg P$ ) and then prove the other ( $Q$ ):

$$P \vee Q \equiv \neg P \implies Q$$

- **Definition 4: Proof by Cases**

If we know that  $P \vee Q$  is true, and we can prove  $R$  from  $P$ , and also prove  $R$  from  $Q$ , we may conclude  $R$ :

- **Example 4.1**

Suppose a natural number is either even or odd, and we want to prove that  $n^2$  is even. We divide the argument into two cases: if  $n$  is even,  $n^2$  is even; if  $n$  is odd,  $n^2$  is even. Since one of these cases must be true,  $n^2$  is even in either case.

- **Example 4.2**

Proof:  $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$

- **Solution 4.2.1**

We want to prove  $\forall a, b, c \in \mathbb{R}, \min\{a, \max\{b, c\}\} = \max\{\min\{a, b\}, \min\{a, c\}\}$ . Let

$$M_1 := \min\{a, \max\{b, c\}\}$$

$$M_2 := \max\{\min\{a, b\}, \min\{a, c\}\}$$

we want to prove  $M_1 \leq M_2$  and  $M_2 \leq M_1$ .

For the case  $M_1 \leq M_2$ , by the definition of  $\min$  and  $\max$

$$\begin{aligned} M_1 &\leq a \wedge M_1 \leq \max\{b, c\} \\ \Rightarrow M_1 &\leq a \wedge (M_1 \leq b \vee M_1 \leq c) \\ \Rightarrow (M_1 &\leq a \wedge M_1 \leq b) \vee (M_1 \leq a \wedge M_1 \leq c) \\ \Rightarrow M_1 &\leq \min\{a, b\} \vee M_1 \leq \min\{a, c\} \\ \Rightarrow M_1 &\leq \max\{\min\{a, b\}, \min\{a, c\}\} \\ \Rightarrow M_1 &\leq M_2 \end{aligned}$$

For the case  $M_2 \leq M_1$ , by the definition of  $\min$  and  $\max$ , we have

$$\min\{a, b\} \leq a, \min\{a, b\} \leq b \leq \max\{b, c\}$$

then  $\min\{a, b\} \leq \min\{a, \max\{b, c\}\}$ .

The other inequality is almost the same. Then  $M_2 = \max\{\min\{a, b\}, \min\{a, c\}\}$ .

- **Example 4.3**

Let  $a_1, a_2, \dots, a_7 \in \mathbb{Z}$ , prove that among these seven integers, one can always choose four numbers whose sum is even.

- **Solution 4.3.1**

Suppose  $a_1, a_2, \dots, a_7 \in \mathbb{Z}$ . Let  $K$  be the number of odd integers among  $a_1, a_2, \dots, a_7$ .

If  $4 \leq K \leq 7$ , then take 4 of these odd integers, their sum is even:

$$(2a + 1) + (2b + 1) + (2c + 1) + (2d + 1) = 2(a + b + c + d + 2)$$

If  $0 \leq K < 4$ , then take 4 of these even integers, their sum is even:

$$2a + 2b + 2c + 2d = 2(a + b + c + d)$$

In both cases, we obtain that we can always choose four numbers whose sum is even.

- **Example 4.4**

Prove that  $\forall n \in \mathbb{Z}$ , if  $n$  is odd, then  $3n$  is odd.

- **Solution 4.4.1**

Suppose  $n \in \mathbb{Z}$ , and  $n$  is odd. Then, there exists an integer  $k \in \mathbb{Z}$  such that  $n = 2k + 1$ .

Then

$$\begin{aligned} 3n &= 3(2k + 1) \\ &= 6k + 3 \\ &= 2(3k + 1) + 1 \end{aligned}$$

Since  $3k + 1$  is an integer,  $3n$  is odd.

◦ **Example 4.5**

For every  $a, b, c \in \mathbb{Z}$ , prove that if  $a \mid b$  and  $b \mid c$ , then  $a \mid c$ .

▪ **Solution 4.5.1**

Suppose  $a, b, c \in \mathbb{Z}$  and  $a \mid b \wedge b \mid c$ . Then there exist  $k, q \in \mathbb{Z}$  such that  $b = ka$  and  $c = qb$ . Then  $c = qka$ , and since  $qk \in \mathbb{Z}$ , by definition,  $a \mid c$ .

◦ **Example 4.6**

Let  $n \in \mathbb{Z}$ . Prove that if  $n^2$  is even, then  $n$  is even.

▪ **Solution 4.6.1**

We will prove the contrapositive:  $\forall n \in \mathbb{Z}$ , if  $n$  is odd, then  $n^2$  is odd.

Suppose  $n \in \mathbb{Z}$  is odd. Then  $\exists k \in \mathbb{Z}$  such that  $n = 2k + 1$ . Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$$

is odd because  $2k^2 + 2k$  is an integer.

◦ **Example 4.7**

Prove that there is no smallest positive rational number.

▪ **Solution 4.7.1**

By contradiction, suppose there exists a smallest positive rational number  $r$ . Then  $\frac{r}{2}$  is a smaller positive rational number. This is a contradiction because  $r$  was assumed to be the smallest. Thus, there doesn't exist a smallest positive rational number.

◦ **Example 4.8**

$\forall \epsilon > 0$ , prove that  $\exists N \in \mathbb{N}$  such that  $\forall n \geq N, \frac{1}{n} < \epsilon$ .

▪ **Solution 4.8.1**

Suppose  $\epsilon > 0$ . We will first prove that  $\exists N \in \mathbb{N}$  such that  $\frac{1}{N} < \epsilon$ .

The Archimedean Property states that for any real number  $x \in \mathbb{R}$ , there exists a natural number  $N \in \mathbb{N}$  such that  $N > x$ . By setting  $x = \frac{1}{\epsilon}$  in the Archimedean Property, we guarantee that  $\exists N \in \mathbb{N}$  satisfying  $N > \frac{1}{\epsilon}$ , and then  $\epsilon > \frac{1}{N}$ . Then if  $n \geq N, \frac{1}{N} \geq \frac{1}{n}$  and  $\frac{1}{n} < \epsilon$ .

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• **Definition 5: Direct Strategy for Implications ( $P \implies Q$ )**

To prove a statement of the form  $P \implies Q$ , we must show that whenever  $P$  is true,  $Q$  is also true.

◦ **Example 5.1**

To prove that if  $n$  is even, then  $n + 1$  is odd, we assume that  $n$  is even and then show that  $n + 1$  is odd.

◦ **Modus Ponens**

If we establish both  $P \implies Q$  and  $P$ , we can conclude  $Q$ .

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• **Definition 6: Indirect Strategy (Proof by Contrapositive)**

Since  $P \implies Q \equiv \neg Q \implies \neg P$ , we can prove  $\neg Q \implies \neg P$  instead of  $P \implies Q$ .

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• **Definition 7: Proof by Contradiction**

To prove  $P \implies Q$ , suppose that  $P$  is true and  $Q$  is false ( $P \wedge \neg Q$ ) and show that this assumption leads to a contradiction ( $R \wedge \neg R$ ):

$$P \implies Q \equiv (P \wedge \neg Q) \implies (R \wedge \neg R)$$

We have:

| $P$ | $Q$ | $R$ | $\neg Q$ | $P \wedge \neg Q$ | $R \wedge \neg R$ | $P \implies Q$ | $(P \wedge \neg Q) \implies (R \wedge \neg R)$ |
|-----|-----|-----|----------|-------------------|-------------------|----------------|------------------------------------------------|
| 1   | 1   | 1   | 0        | 0                 | 0                 | 1              | 1                                              |
| 1   | 1   | 0   | 0        | 0                 | 0                 | 1              | 1                                              |
| 1   | 0   | 1   | 1        | 1                 | 0                 | 0              | 0                                              |
| 1   | 0   | 0   | 1        | 1                 | 0                 | 0              | 0                                              |
| 0   | 1   | 1   | 0        | 0                 | 0                 | 1              | 1                                              |
| 0   | 1   | 0   | 0        | 0                 | 0                 | 1              | 1                                              |
| 0   | 0   | 1   | 1        | 0                 | 0                 | 1              | 1                                              |
| 0   | 0   | 0   | 1        | 0                 | 0                 | 1              | 1                                              |

- **Definition 8: Disproofs and Counterexamples**

A statement is disproved if we show a single instance in which it is false, known as a **counterexample**.

For a universally quantified statement  $\forall x P(x)$ , a single object  $x_0$  for which  $P(x_0)$  is false suffices to disprove the statement.

For an implication  $P \implies Q$ , a counterexample consists of a situation where  $P$  is true and  $Q$  is false.

- **Example 8.1**

Consider the statement  $x^2 - 1 > 0$  for all  $x \in \mathbb{R}$ . To disprove it, use the counterexample  $x_0 = 1$ , which gives  $1^2 - 1 = 0 \not> 0$ . Thus, the statement is false.

- **Example 8.2**

Prove that the existentially quantified statement  $(\exists x \in \mathbb{R}) \left( \frac{1}{x^2+1} > 1 \right)$  is false.

- **Solution 8.2.1**

- We need to show that  $\frac{1}{x^2+1} \leq 1$  for every  $x \in \mathbb{R}$ . Since  $0 \leq x^2$  for all  $x \in \mathbb{R}$ , it follows that  $1 \leq x^2 + 1$ . Because  $x^2 + 1 > 0$ , taking the reciprocal reverses the inequality:

$$\frac{1}{x^2 + 1} \leq 1$$

for every  $x \in \mathbb{R}$ . Thus, the original statement is false.